

Tristan Hafer

Desktop Application Developer

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SUMMARY

Systems-level developer experienced in building and shipping cross-platform Windows services and CLI tooling, with hands-on work in cross-compilation, native installer packaging, and CI/CD pipelines for desktop software — alongside six years of broader software engineering experience in Rust, C#, and C/C++.

SKILLS

Languages: Rust, C#, Zig, C/C++ **Platforms:** Cross-platform Windows service/CLI development, cross-compilation (Zig + Cargo, x86_64-pc-windows-gnu) **Packaging & Deployment:** WiX/wixl/msitools Windows installer generation (from Linux), self-hosted GitLab CI/CD **Systems:** Linux administration, Docker, Windows service architecture

EXPERIENCE

Applied Cardiac Systems, Inc. — Irvine, CA *Project Manager, Software Developer* — September 2019 – Present

"geoff" — cross-platform Windows service/CLI tool

- Built and maintain a cross-platform Windows service and CLI tool
- Configured a `build.zig` to cross-compile the tool from Rust/Cargo to the `x86_64-pc-windows-gnu` target using Zig
- Built and debugged a GitLab CI pipeline that generates Windows installers using WiX tooling (`wixl/msitools`) running on Linux, including tracking down a Debian trixie packaging regression that dropped the `wixl` binary

Core 2/3 — Cat M1-enabled ambulatory heart monitor | 2021 – Present

- Developed C/C++ firmware for the device software layer of a real-time ECG monitoring and remote alerting system, including power management and transmission protocol optimization (see Embedded Systems resume for full detail)

Intranet — engineering support tooling | 2021 – Present

- Designed and deployed an internal Docker-based suite for GitLab project management, CI/CD, VPN, and DNS, supporting desktop and service software delivery

REFERENCES

Available upon request — including Wes Koerber (Sr. Software Developer) and Braden Walker (Sr. Full-Stack Web Developer)